

VISHNU K

Senior Frontend Engineer | React.js & Next.js

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🐙 Github 📁 Portfolio

PROFILE

Results-oriented Frontend Engineer with **4 years of experience** at **Tata Consultancy Services**, specializing in building high-performance web applications using **React.js, Next.js, and TypeScript**. Proficient in **Micro-frontend architecture**, performance optimization (Core Web Vitals), and scalable UI design. Recently expanded technical capabilities into Backend development (Node.js, Express, MongoDB) to deliver full-stack solutions. Proven track record of improving SEO, reducing build times by 25%, and delivering seamless e-commerce experiences

SKILLS

- **Core Frontend:** React.js, Next.js (SSR/SSG), React Native, TypeScript, JavaScript (ES6+), HTML5, CSS3.
- **Backend (MERN):** Node.js, Express.js, MongoDB, RESTful APIs
- **Architecture & Tools:** Micro-frontends (Module Federation), Git
- **Testing & Performance:** Jest, Lazy Loading, Memoization
- **UI Libraries:** Material UI, Styled Components.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services,

2022 Jan – present | Kochi, India

System Engineer (Frontend Developer) 📄

- **E-Commerce Platform Migration & Modernization:**
 - Spearheaded the migration of legacy e-commerce flows to a modern **Next.js + React** architecture, resulting in a **40% improvement in page load speed** and significantly better SEO rankings.
 - Designed and implemented **Micro-frontend architecture** using Module Federation; this decoupled the codebase, enabled independent deployments, and reduced build times by **25%**.
- **Feature Development & Optimization:**
 - Engineered core user-journey features including *Express Checkout*, *Add to Cart*, and an AI-driven *Chatbot*, directly contributing to an increase in user retention and conversion rates.
 - Optimized **Product Listing Pages (PLP)** to handle large datasets efficiently using virtualization and lazy loading, ensuring 60FPS scrolling performance.
- **Scalability & Maintenance:**
 - Built a shared internal UI Component Library using **TypeScript and Material UI**, reducing code duplication by 30% across multiple squads.
 - Integrated frontend applications with **AEM (Adobe Experience Manager)**, enabling marketing teams to update content dynamically without developer intervention.

- Led code reviews and mentored junior associates on React best practices and state management strategies.

PROJECTS

Tractor Supply Co. (Client Project),

2024 Jun – present

Tech Stack: Next.js, TypeScript, AEM, Module Federation

- Delivered a high-traffic e-commerce catalog page. Focused on Server-Side Rendering (SSR) to optimize for Core Web Vitals and SEO.
- Project URL: <https://www.tractorsupply.com/tsc/catalog/dog-food> 

MERN Attendance System (Full Stack)| Personal

Portfolio Project, *Tech Stack: MongoDB, Express.js,*

React.js, Node.js, Material UI 

- Designed a full-stack application to digitize classroom attendance tracking.
- Developed secure **RESTful APIs** with Node.js and Express
- Implemented complex MongoDB aggregations to generate daily and monthly attendance reports.
- *Outcome:* Deployed successfully on Render/Netlify and currently used for beta testing.

MICHAEL'S, *React Native | API Integration | UI Development*

- Developed complete store app UI and core flows.

SHOWTIME - Movie Discovery App,

Tech Stack: React Native, TMDB API, Axios

- Built a cross-platform mobile application featuring real-time movie data, trailer integration via YouTube API, and infinite scrolling.
- **GitHub:** [SHOWTIME](#)  | **APK:** [SHOWTIME](#) 

EDUCATION

Bachelor of Technology

Electrical and Electronics Engineering

Rajiv Gandhi Institute of Technology,Kottayam